

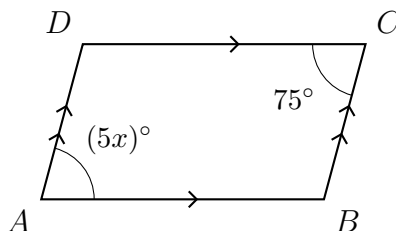
Classifying Quadrilaterals from Defining Properties

Naming the most specific quadrilateral from the marks on a figure, classifying from what the diagonals do, and classifying from vertex coordinates

Directions.

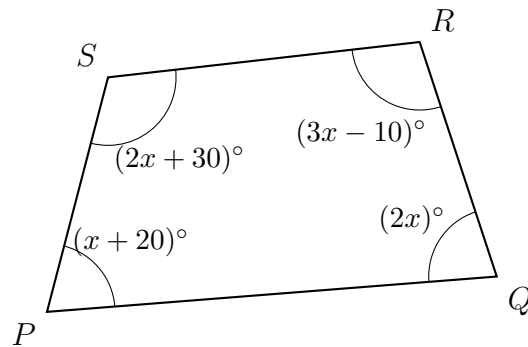
- Every problem asks for the *most specific* name the given properties force — “parallelogram” is wrong when the marks already force “rhombus”.
- Classify from the marks and the stated facts, never from how the drawing looks; some figures are labelled *not to scale* for that reason.
- Then use *that* family’s defining property to set up the equation, and write the property you used beside your work.
- Matching ticks mean congruent, matching chevrons mean parallel, and a small square means a right angle.

1. The chevrons show $\overline{AB} \parallel \overline{DC}$ and $\overline{BC} \parallel \overline{AD}$. Name the most specific quadrilateral these marks force, then use that family’s angle property to find x .



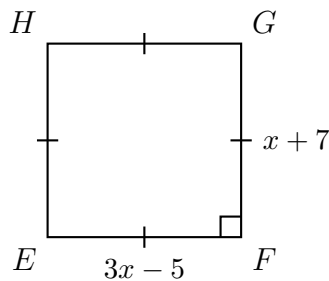
Answer: _____

- 2.** This figure carries no congruence marks and no parallel marks at all. Say what that means about naming it, then find x using the one angle fact that holds for *every* quadrilateral. (*Not to scale.*)



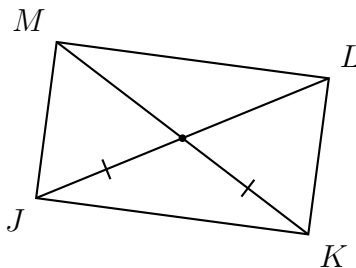
Answer: _____

- 3.** All four sides carry the same single tick, and one vertex carries a right-angle mark. Name the most specific quadrilateral, then use its side property to find x .



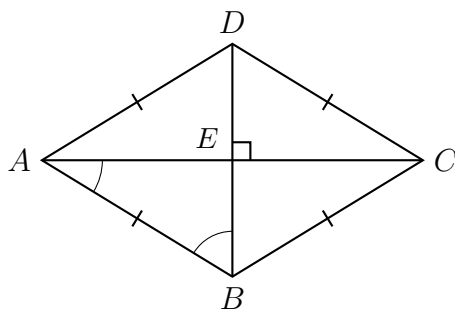
Answer: _____

4. In quadrilateral $JKLM$ the diagonals $\overline{JL}$ and $\overline{KM}$ bisect each other, and the single ticks show that they are also congruent. Their lengths are $JL = 4x - 6$ and $KM = x + 9$. Name the most specific quadrilateral those two diagonal facts force, then find x and the length JL . (*Not to scale.*)



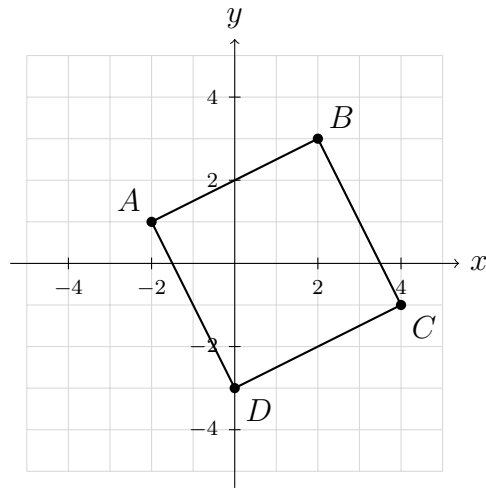
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5. In quadrilateral $ABCD$ all four sides are congruent and the diagonals meet at E at a right angle, each bisecting the other, with $m\angle BAE = (2x)^\circ$ and $m\angle ABE = (4x - 6)^\circ$. Name the most specific quadrilateral, then use the right angle at E to find x .



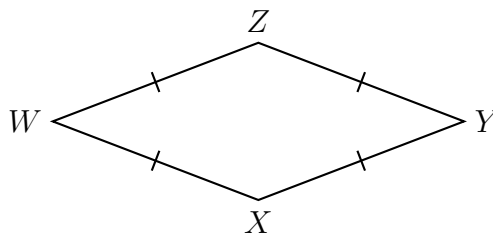
Answer: _____

6. Quadrilateral $ABCD$ has vertices $A(-2, 1)$, $B(2, 3)$, $C(4, -1)$ and $D(0, -3)$. Find the slope of $\overline{AB}$, the slope of $\overline{BC}$, and the lengths AB and BC (round the lengths to two decimal places). Then give the most specific name for $ABCD$, saying which computed fact forces each part of that name.



Answer: _____

7. Marcus says “every rhombus is a rectangle, because both of them are parallelograms.” Decide whether that is *always*, *sometimes* or *never* true. Justify your choice by naming the defining property that separates the two families, then mark on the figure below what would have to be added to force this rhombus to be a rectangle as well.



8. In quadrilateral $QRST$ the marks show only that $\overline{QR} \cong \overline{TS}$ and $\overline{QR} \parallel \overline{TS}$ — one pair of opposite sides, congruent *and* parallel. Write an argument that $QRST$ must be a parallelogram, naming the property or theorem at each step. Then state whether these marks also force $QRST$ to be a rectangle, and say why.

